

Linux APT Package Management Lab

Summary: In this lab, I worked in a Debian-based Linux environment to practice installing and managing security tools using the Advanced Package Tool (APT) package manager. My role was to act as a security analyst responsible for ensuring that Suricata and tcpdump were properly installed on the system using `sudo` and APT. The goal was to practice basic Linux package management and confirm that both tools were installed and functioning as expected.

Commands Used:

```
apt
sudo apt install
suricata
sudo apt remove
apt list --installed
```

Overview: I started by running the `apt` command on its own. When APT is installed, running that command prints usage information. This confirmed that the package manager was available and working on the system. Next, I installed Suricata using `sudo apt install suricata`. The `sudo` command lets me run administrative tasks with elevated privileges, which is required for installing or removing software. The `apt install` portion tells the package manager to download Suricata and any dependencies it needs.

After installation, I ran `suricata` by itself to verify that the application was installed correctly. When a program is installed, running its name usually prints version or usage information, which confirmed that Suricata was installed successfully. To simulate a troubleshooting scenario, I removed Suricata using `sudo apt remove suricata`. This command removes the package but leaves the configuration files behind. Running `suricata` again returned an error, which confirmed that the binary was no longer present.

I then installed tcpdump using `sudo apt install tcpdump`. Once the installation finished, I used `apt list --installed` to view all installed packages. This command prints a long list of everything currently installed on the system, and I used it to confirm that tcpdump appeared in the list. Finally, I reinstalled Suricata using the same install command as before so that both tools were available on the system.

Reflection: This lab reinforced the basics of managing software on a Linux system. Using `sudo` and APT is something I'll be doing regularly as a security analyst, especially when installing or troubleshooting security tools. Even though the tasks were simple, they helped me get more comfortable verifying installations, checking dependencies, and understanding how Linux handles packages. These are foundational skills that support more advanced work